Healthy Food for Healthy Peoples

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***Abstract*—Diseases associated with lifestyle choices are becoming a serious threat to global health; these include obesity, diabetes, high blood pressure, and heart disease. The main cause of these problems is the growing reliance on highly processed and convenience foods. People are also becoming less educated on nutrition and healthy food choices. Nutrition-related information is widely available on the internet, and this extends to information on dietary approaches. However, the majority of this information is too complex for the average person, and is written at a level that can be difficult to understand. Additionally, creating a nutrition plan that is not only practical, but that can also be integrated into daily life is challenging for many people. This creates a clear need for dietary support systems. This is the problem Healthy Food for Healthy People set out to solve in an innovative way. The team intends to use online approaches to create a personalized diet suggestion system that is based on cutting-edge nutrition science and innovative data-driven nutrition decision systems. The system will assess a user's caloric needs and provide personalized meal suggestions based on inputs related to the user's age, height, weight, BMI, food preferences, and health goals. The Smart Thali System is designed for use in India, and is designed to provide meals that are balanced in their inclusion of carbohydrates, proteins, fats, vitamins, and minerals.**

Unlike most diet apps that promote calorie counting, this system focuses on balanced nutrition, culturally preferred foods, and encourages the use of foods that can be found in the local market.

***Index Terms*—Personalized Diet Planning, Smart Thali, Nutrition Informatics.**

I. Introduction

Diseases such as obesity, diabetes, and heart disease are telling us that we need to be more concerned with how we eat. Not to mention the fact that we need to start eating regularly and need to be more aware of the nutrients that we are consuming. There is a lot of information out there that is associated with diets but many people struggle with the concept of creating a daily balanced meal plan that adheres to the guidelines of the diet. A lot of diet apps are western-centered, meaning that they only allow the consumption of certain foods associated with 'diets' and therefore don't fit the eating habits of an Indian consumer. A traditional Indian Thali is ideal as it is designed to be a balanced meal. However, the Thali portion method is not clear to consumers using the Thali. This is what the project Healthy Food for Healthy People aims to achieve with its online personalized diet recommendation system. The system provides consumers with the means to input their age, weight, and any other relevant information to estimate their daily nutritional needs and create a Smart Thali to help them with their diet. The system provides users with access to basic nutritional analytics and a limited amount of data to help them better understand the system It aims to achieve a better system. The goal of the system is to improve healthier consumption of food and habits through a simple Thali system that incorporates various technological processes.

II. Problem Statement

Recent changes in lifestyle have caused significant changes in dietary habits and have led to an increase in fast food intake as well as eating irregularity and a decrease in nutritional knowledge. The knowledge to create a meal that has a combination of an adequate amount of protein, carbohydrates, fat and vitamins is missing in many people. This has led to an increase in lifestyle related diseases in numerous age groups. Although a plethora of work out and diet apps exist, an over whelming amount of these apps have a westernized food, count the calories, and diet chart that a typical Indian could never follow. Most of these apps charge a subscription, track everything, and require a knowledge of nutrition that most people, especially students, do not have. The Indian "Thali" has a tradition that is culturally accepted as a means of an entire balanced meal however the method of construction of a Thali is not fully understood by the majority of the population in relation to a person's age, body weight, and nutritional requirements. This justifies the need for a system that is culturally accepted, simple, and easy to use guidance for users to create a balanced meal Indian meal. This is the reason, in this project, an interactive and simple platform is created to assist users in understanding nutritional information, generating balanced meal ideas, and creating a web based solution to aid in the creation of a balanced meal and promoting good eating habits.

III. Literature Review

As obesity, diabetes, and cardiovascular disease continue to rise, the importance of proper nutrition and diet management has come to the forefront. Over the years, a plethora of digital approaches such as mobile apps, websites and smart recommender systems have been developed to assist individuals in monitoring their diet and making healthier eating choices. In this paper, we discuss the advancements made in nutrition informatics, personalized nutrition recommendation systems and technology-enabled health support systems. Historically, diet management systems have been centered around a summative approach to calorie counting and food logging. Calorie tracker

apps were developed to allow users to log their meals and keep track of their calorie intake. Though these systems created a heightened awareness around calorie intake, they were not equipped with personalized mechanisms and required tedious manual entries. This, in addition to the weak cultural flexibility and social support, meant that many users dropped the systems after a few weeks. Due to advancements in data analytics and AI, the personalization features have been far easier to implement. Smart food suggestion systems have aimed to personalize food recommendations based on the users age, weight, height, activity level, diet preferences, and Body Mass Index (BMI).

IV. OBJECTIVE

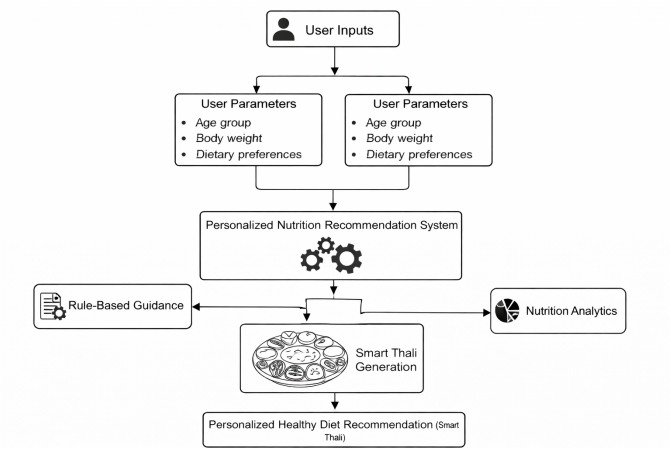
- Main Aim:
  - The main aim of the project is to create a web-based personalized nutrition recommendation system that incorporates the Smart Thali concept to give users meal suggestions that have balanced nutrition and promote healthy eating.
- Secondary Aims:

1) User Data Collection and Profiling
   - Gathering of the necessary user parameters like age group, body weight, eating habits, and health objectives in order to develop an individual nutrition profile.
2) Nutritional Requirement Estimation
   - Calculate the daily required nutrition intake, which include calories, and the macro nutrient breakdown, (proteins, carbs, and fats) based on the accepted dietary guidelines.
3) Smart Thali Generation
   - The creation of a cohesive Smart Thali design that suggests balanced Indian meal combinations from the food pyramid (cereals, pulses, vegetables, dairy, and others).
4) Nutritional Database Integration
   - To have a well defined database for meal suggestions comprising of the nutritional values of the Indian foods that are consumed on a daily basis.
5) Nutrition Analytics and Visualization
   - Providing users with analysis and graphics (bar charts and others) that aid in the understanding of the distribution of nutrients and the balance of their diet.
6) Rule-Based Recommendation Engine
   - Adopting a recommendation system based on logic and user inputs that is adequate to nutritional directives to give meal recommendations.
7) Awareness and Preventive Healthcare Support to promote awareness of nutrition and preventive healthcare by making informed and balanced food choices.
8) Future Development and Scalability To build the system modularly so that future integrations of backend databases, machine learning, real-time tracking, and personalization at high levels can be added.

V. MARKET SURVEY

According to my research, a part of the digital-gaming boom, the mobile fitness and nutrition app market has rapidly expanded. Health-conscious consumers can now download an app to monitor calories, track meals, calculate BMI, and record physical activity. Most of the popular apps focus on tracking calories consumed and burned in order to help users achieve their weight goals. However, the apps available on the market today have several limitations. First, the majority of mobile fitness and nutrition apps are designed to cater to Western dietary habits. This may not be appropriate for users in India because of their unique eating habits, the availability of different types of foods, and their diverse cultures. Users also find the mobile apps that require them to track calories to be overly complicated, tedious, and unsustainable. Also, for many popular mobile fitness and nutrition apps, premium features that require a paid subscription are inaccessible to students and other users. Most mobile fitness apps also fail to adequately address the user's need to maintain a balanced diet. Users can track their caloric intake through the app, but have the option to eat an unbalanced diet because of the app's lack of focus on the proportions of carbohydrates, proteins, and fats that their diet contains. Very few mobile nutrition and fitness also simplify the user's task of tracking their food intake and educate the users on the fundamentals of food tracking, which adversely affects new users. Very few mobile nutrition and fitness also simplify the user's task of tracking their food intake and educate the users on the fundamentals of food tracking, which adversely affects new users.

Users need a system that suggests meals, and guides them on how to achieve nutritional balance in a simple and actionable way. The system, Healthy Food for Healthy People, focuses on culturally appropriate Indian food patterns, Smart Thali generation and uncomplicated nutritional feedback to address these gaps. It focuses on preventive healthcare and health awareness. It is a free, web-based, easy-to-use solution.



- System Architecture Proposal Explanation The illustration describes how the **Healthy Food for Healthy People** personalised nutrition recommendation system based on Smart Thali works. The system starts with **User Inputs** The users provide basic information including age group, body weight, and food preferences. All information provided is critical in forming the system's decisions. The system then classifies the information under **User Parameters** and converts the data to usable attributes. The system then sends the parameters to the **Personalized Nutrition Recommendation System**, which is the system's main processing unit. This unit, through a combination of provided nutrition guidelines and logic, determines the user's macronutrient requirements and subsequently, how to meet the user's needs through meal suggestions. It ensures that every meal suggested to the user is correctly portioned for carbs, protein, fat, and other nutrients. The system works with the following other 2 modules: **1.Rule-Based Guidance–** This module determines that the suggestions are nutritionally appropriate according to the user's age and weight through a collection of dietary rules and health recommendations. **2.Nutrition Analytics–** analyzes the breakdown of the nutrients present in the food being consumed through graphical representations, in order to understand the user's food intake balance. From this, the system creates a **Smart Thali** which provides the necessary food groups that constitute a well-balanced Indian meal, considering the nutritional value of food. The system also generates a **Personalized Healthy Diet Recommendation** based on the user's profile.

  Flow Summary:

  User Inputs → Parameter Processing → Recommendation Engine → Smart Thali Creation → Personalized Diet Recommendation Output

VI. REFERENCES

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